

PUMPING LEMMA FOR REGULAR SETS

Dr. A. Beulah
AP/CSE

INTRODUCTION

- A Regular language is a formal language that can be expressed using a regular expression
- A regular language satisfies the following equivalent properties:
 - it is the language accepted by a nondeterministic finite automaton
 - it is the language accepted by a deterministic finite automaton
 - it can be generated by a regular grammar
 - it can be generated by a prefix grammar
 - it can be accepted by a read-only Turing machine
- Regular set is a set of strings of a Regular Language
- For every regular language there is a FA that accepts the language

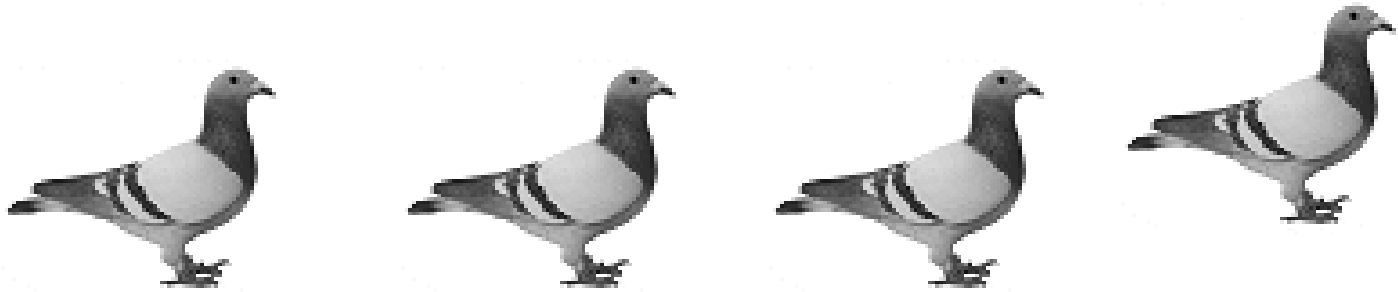
THE PIGEONHOLE PRINCIPLE

- If you put n pigeons into m pigeonholes, and $n > m > 0$, then at least at least two pigeons are in the same pigeonhole.

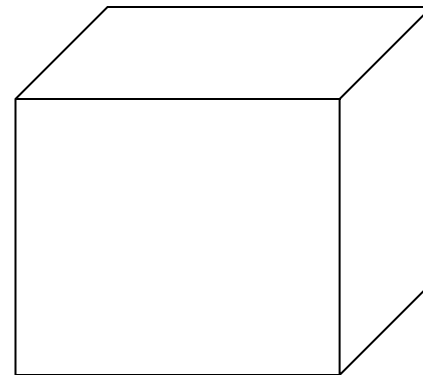
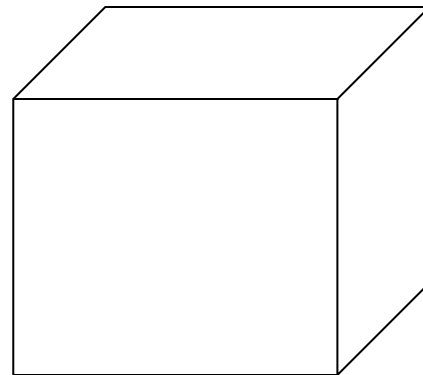
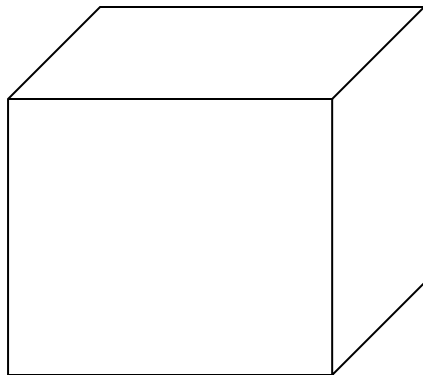


THE PIGEONHOLE PRINCIPLE

4 pigeons

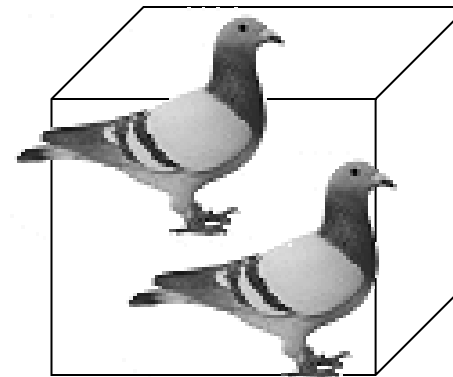
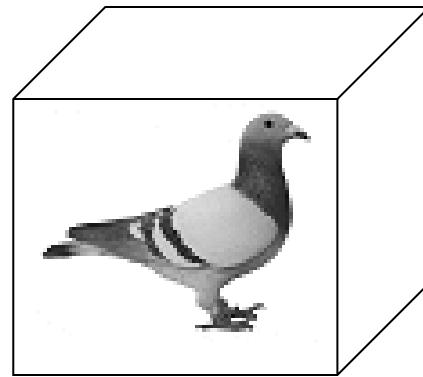
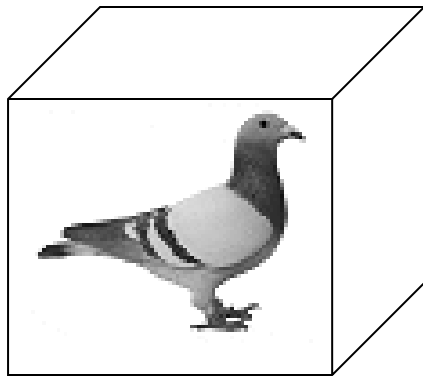


3 pigeonholes



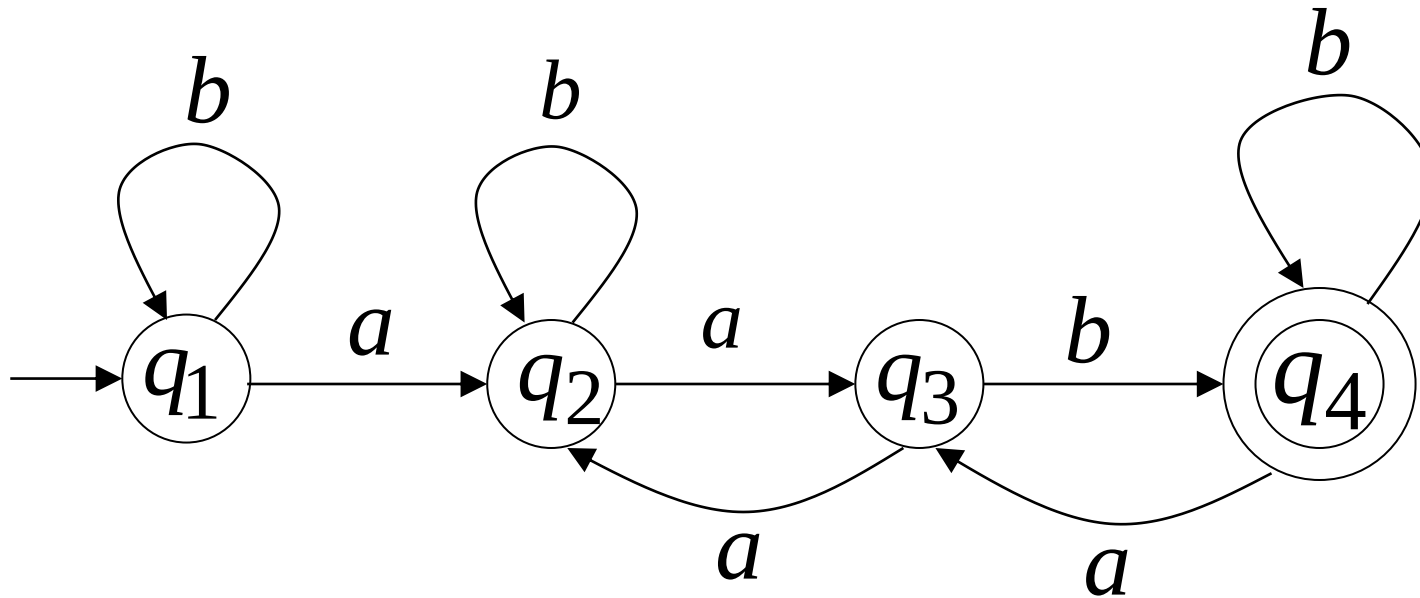
THE PIGEONHOLE PRINCIPLE

A pigeonhole must
contain at least two pigeons



PIGEONHOLE PRINCIPLE & DFAS

Consider a DFA with 4 states

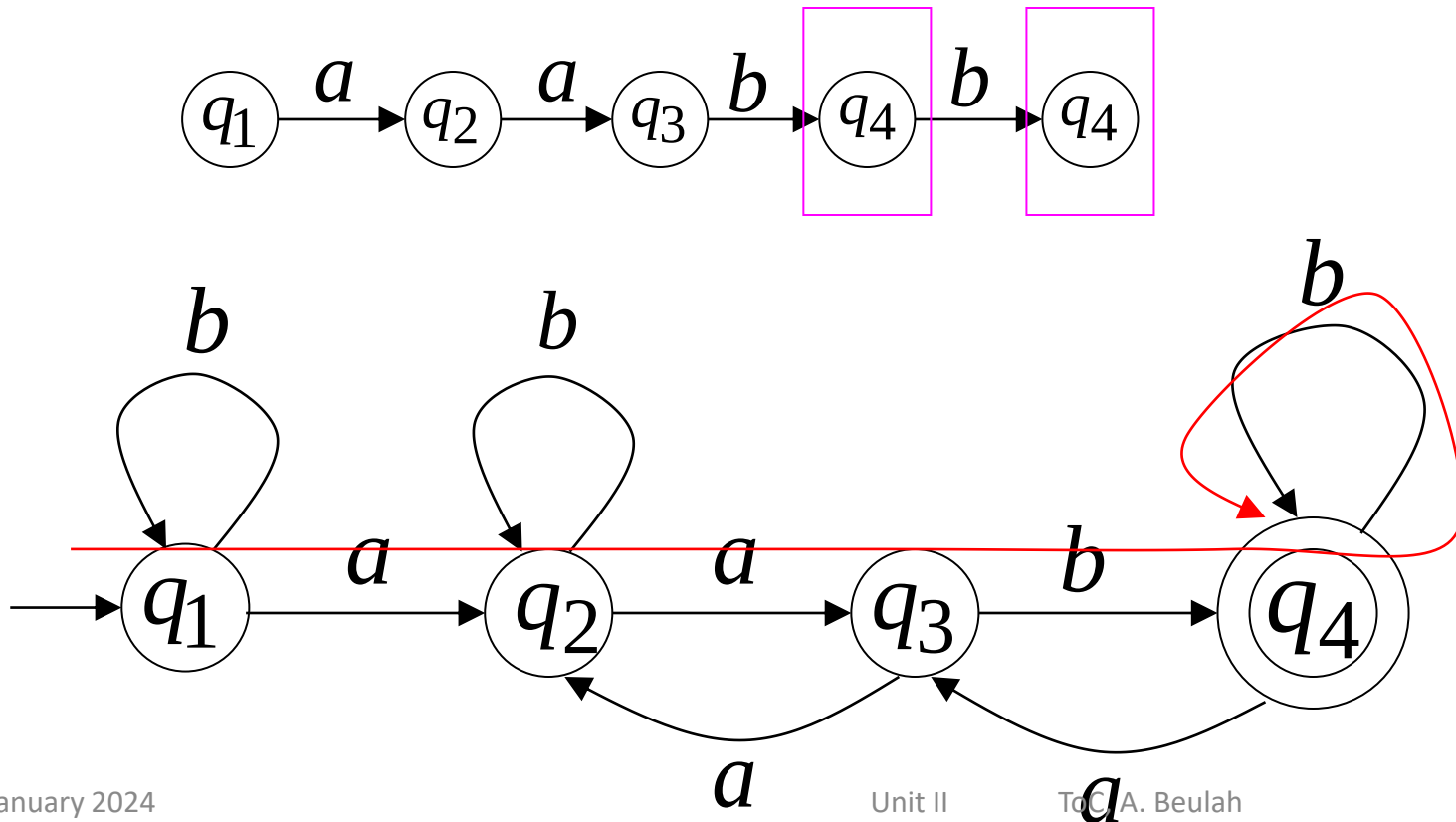


PIGEONHOLE PRINCIPLE & DFAS

- Consider the walk of a “long” string: $aabb$ (≥ 4)

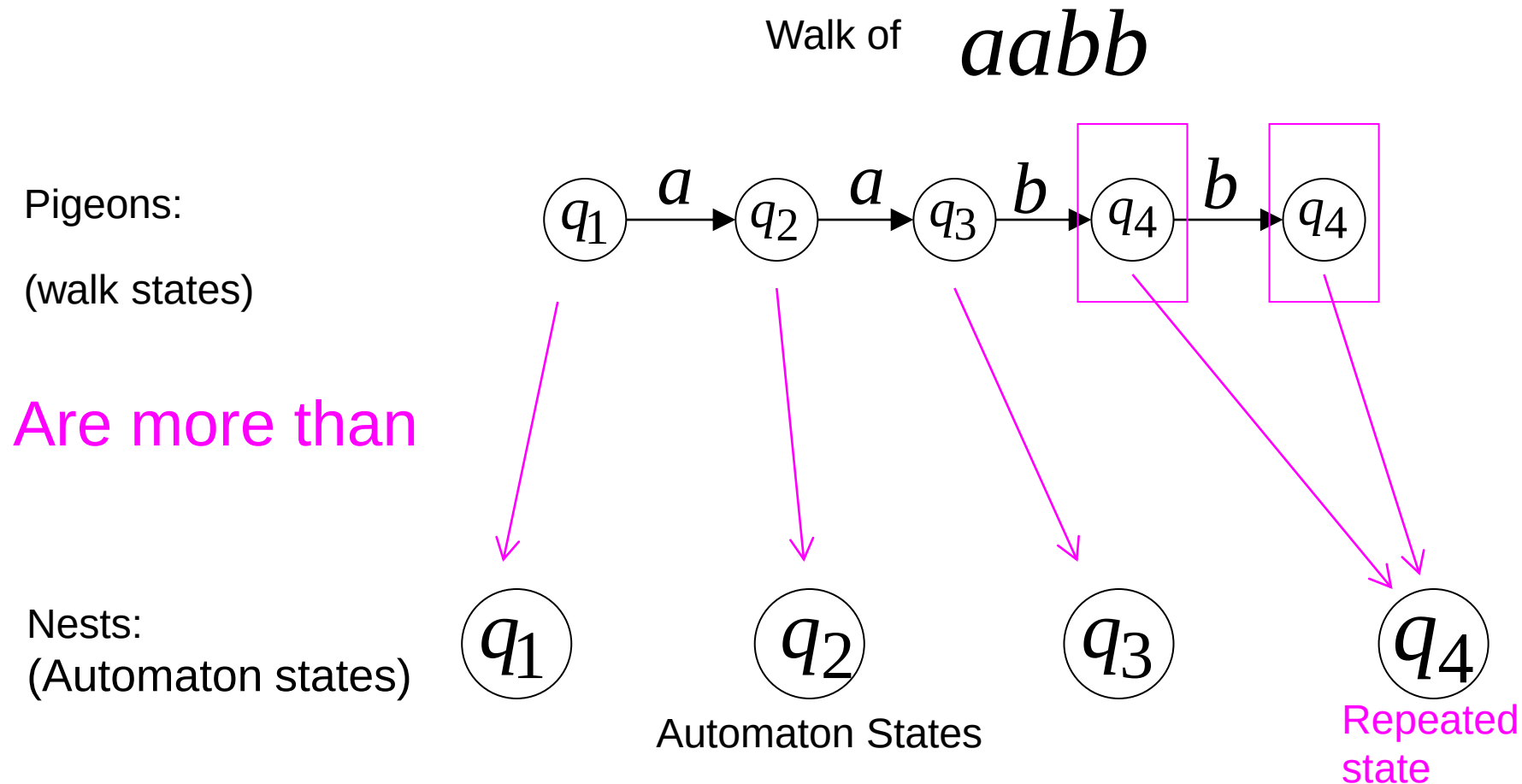
A state is repeated in the walk of

$aabb$



PIGEONHOLE PRINCIPLE & DFAS

- The state is repeated as a result of the pigeonhole principle



PUMPING LEMMA

- Describes an essential property of all regular languages
- For a particular language, any sufficiently long string in the language contains a section, or sections, that can be removed, or repeated any number of times(pumping), with the resulting string remaining in that language
- The pumping lemma is used to prove that a particular language is non-regular

PUMPING LEMMA

- Let L be a regular language. Then there is a constant n (which depends on L / number of states in FA) such that for every string w in L such that $|w| \geq n$, we can break w into three strings, $w = xyz$, such that $y \neq \varepsilon$ ie $|y| > 0$, $|xy| \leq n$, and for all $i \geq 0$, xy^iz is also in L .
- Proof
 - Let n be $|Q|$.
 - If $w \in L$ and $|w| \geq n$. Let $w = a_1 a_2 \dots a_m$, where $m \geq n$.
 - $\delta(q_0, a_1 a_2 \dots a_i) = q_i$, $i = 1, 2, \dots, m$.

$$L = \{ \underline{w} \mid \dots \}$$

$$|w| \geq n$$

$$w = xyz$$

$$(i) |xy| \leq n$$

$$(ii) |y| > 0$$

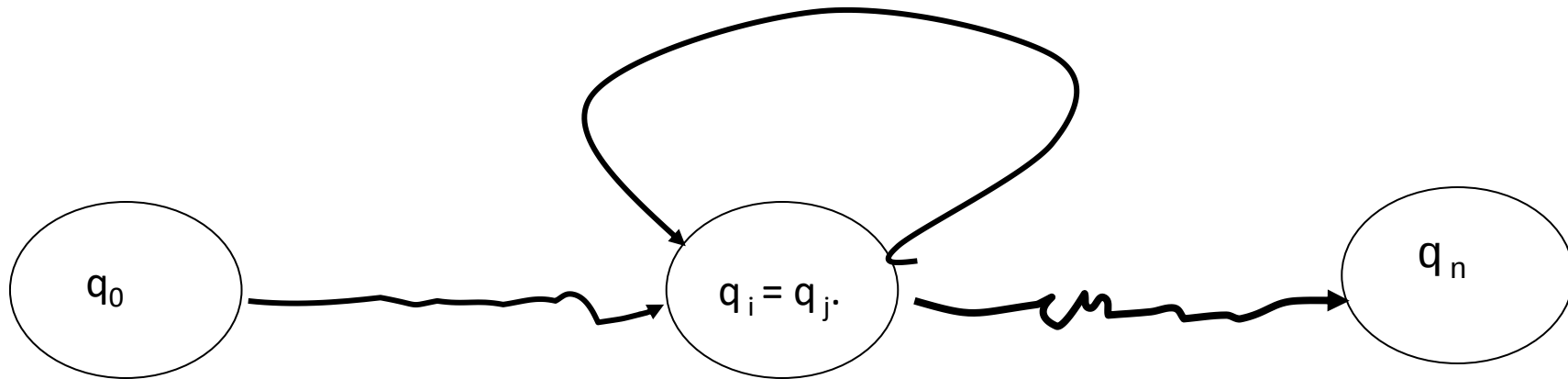
$y \neq \varepsilon$

$$\underline{xy^i z} \quad \forall i \geq 0$$

EL

PUMPING LEMMA

- Since there are only n states in Q and $m \geq n$, by the pigeon hole theorem there are two states of $q_0, q_1, q_2, \dots, \text{ and } q_n$ are same, say $0 \leq i < j \leq n$ and $q_i = q_j$.



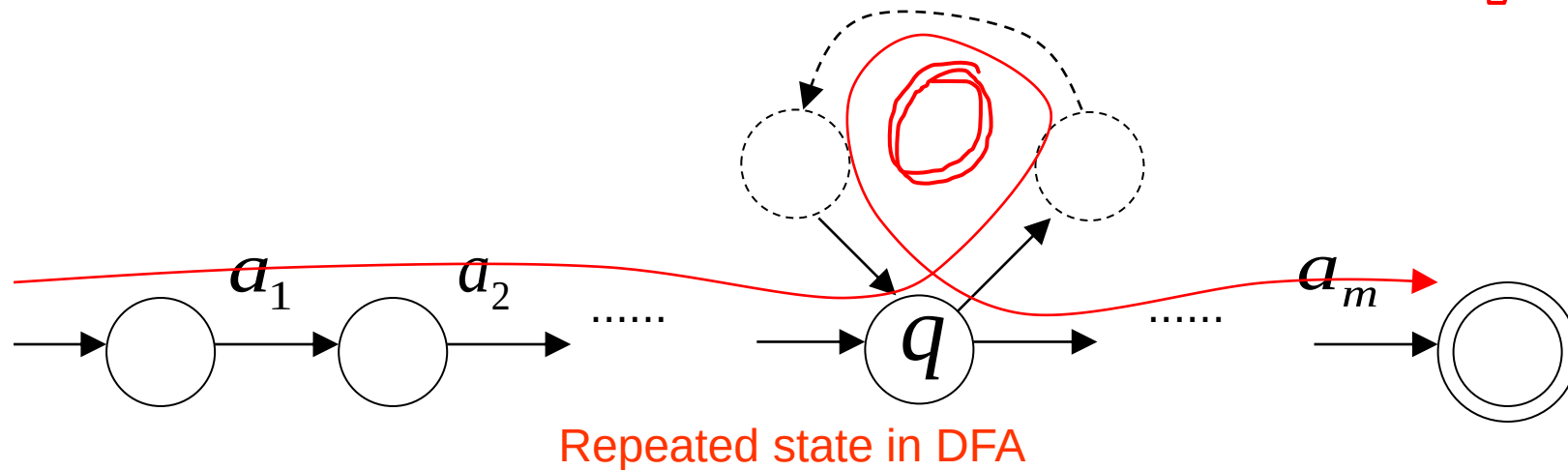
PUMPING LEMMA

- If $w \in L$ and $|w| \geq n$, then, at least one state is repeated in the walk of w

Walk in DFA of

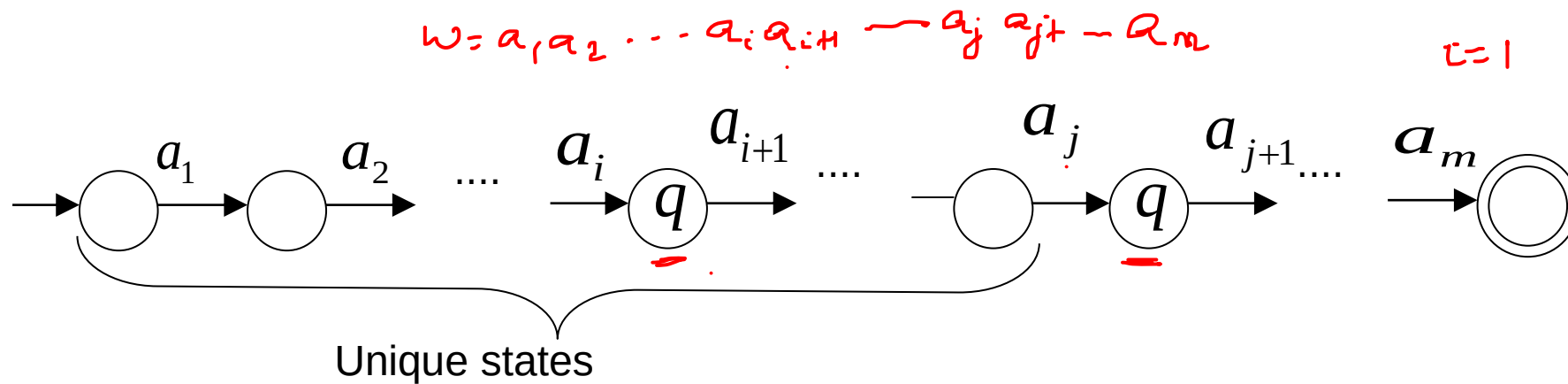
$$w = a_1 a_2 \cdots a_m$$

$xy^2 y^i 0 \dots 0$



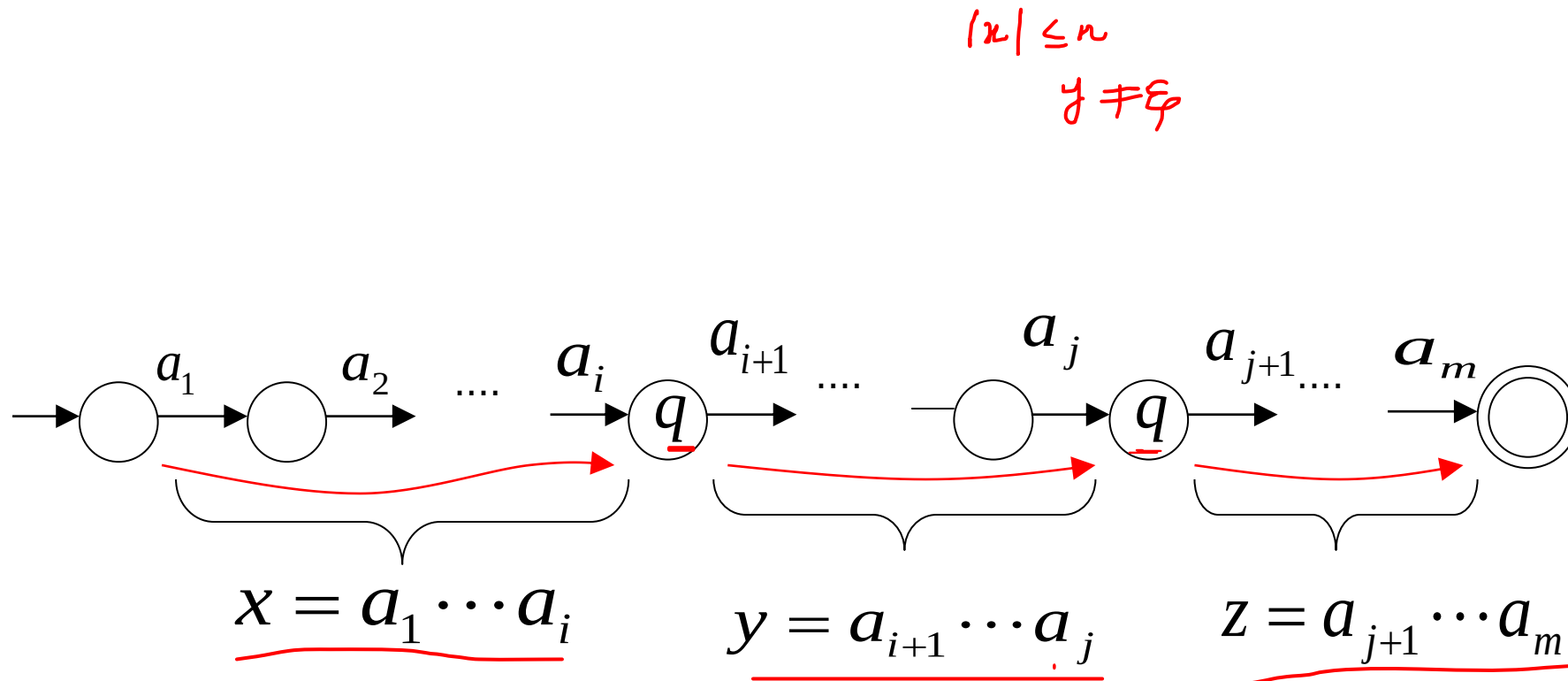
PUMPING LEMMA

- There could be many states repeated
- Take q to be the first state repeated



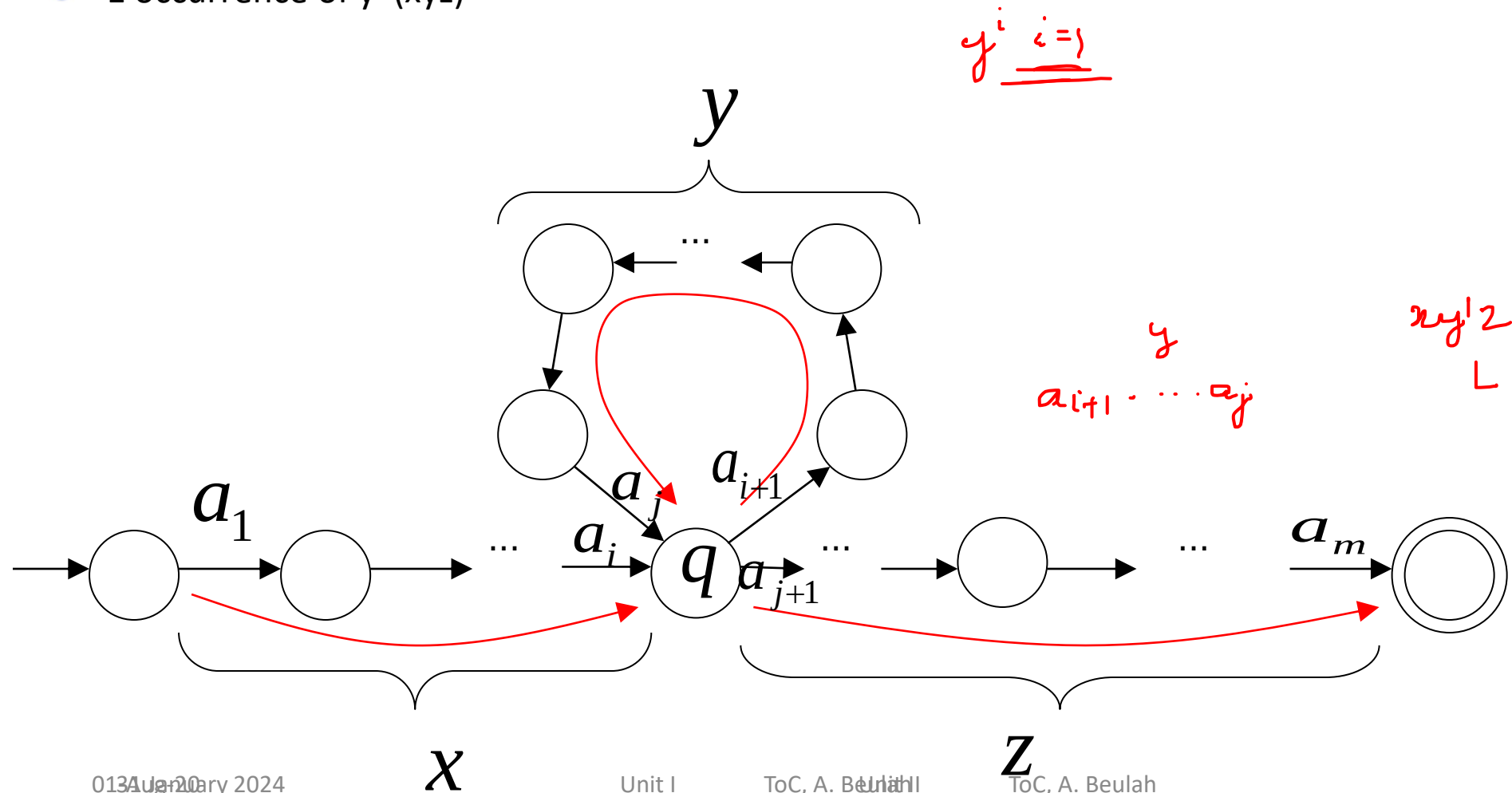
PUMPING LEMMA

- We can write $w = xyz$



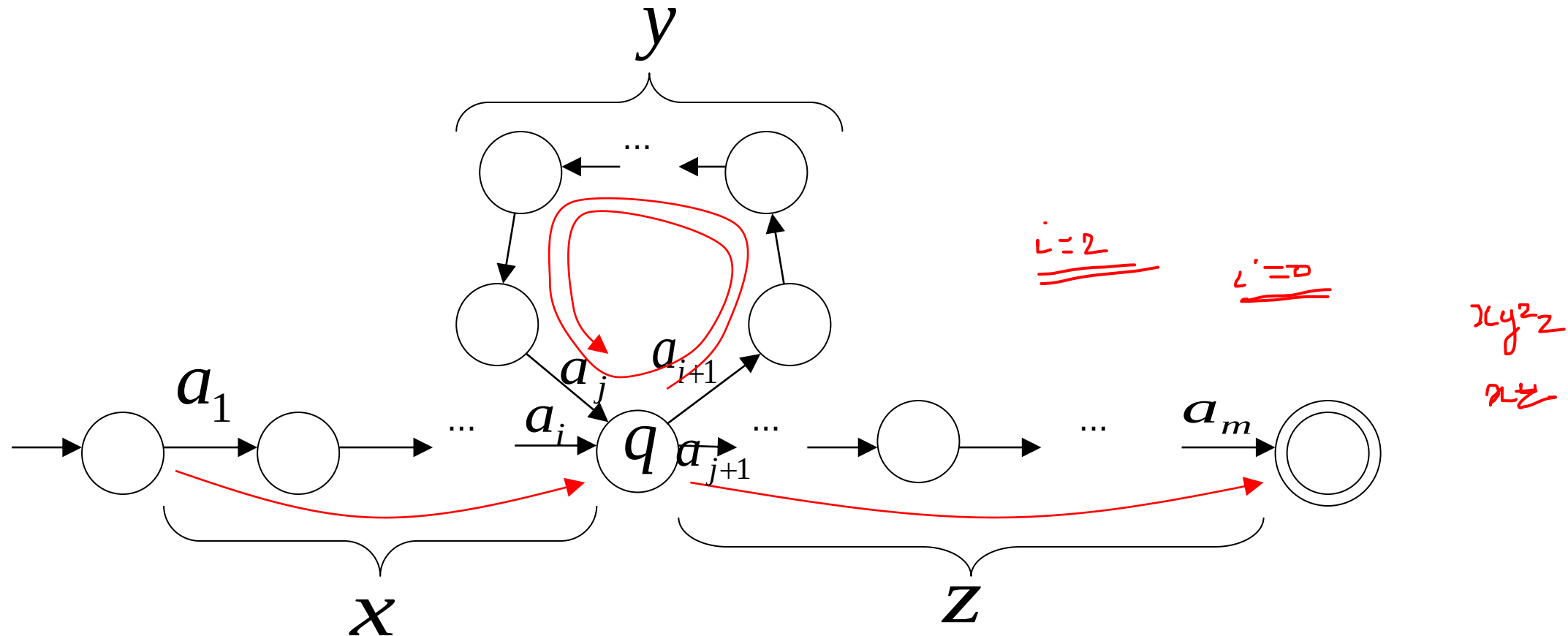
PUMPING LEMMA

- In DFA: write $w = xyz$
- 1 occurrence of y (xyz)



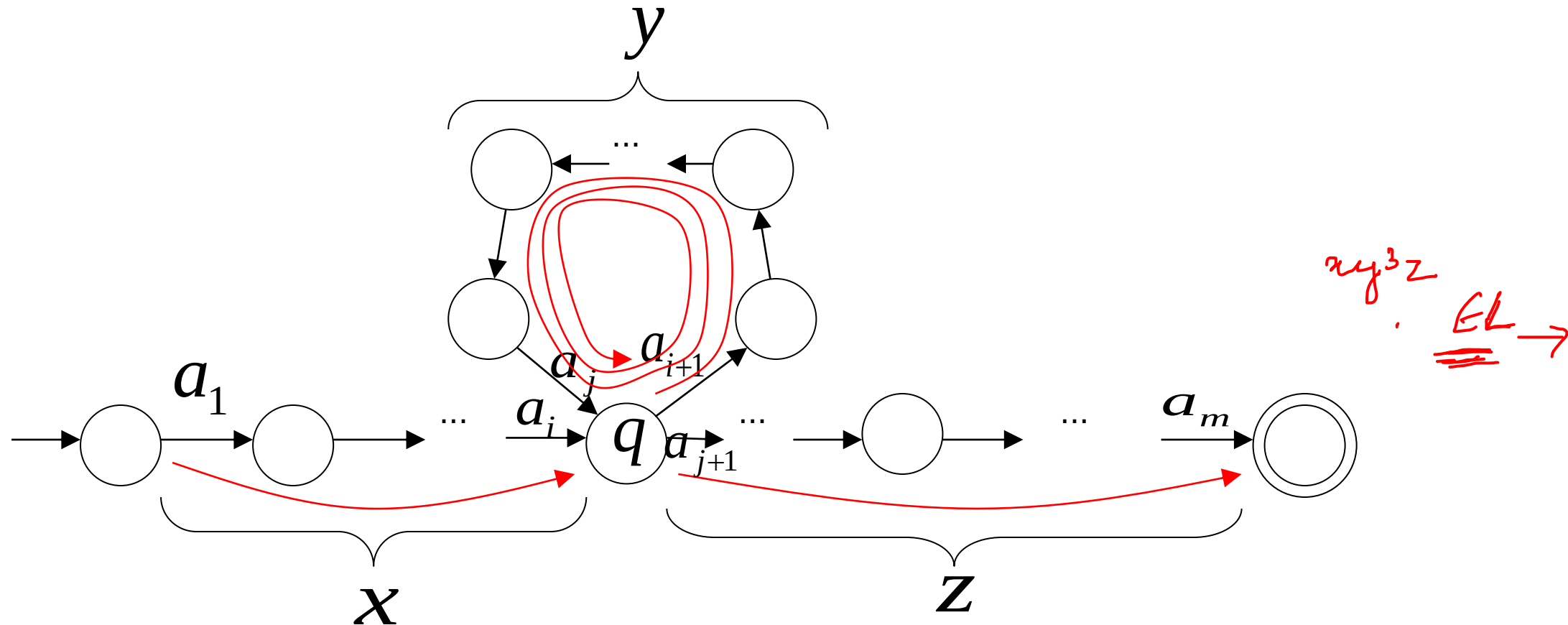
PUMPING LEMMA

- 2 occurrences of y ($xyyz$)



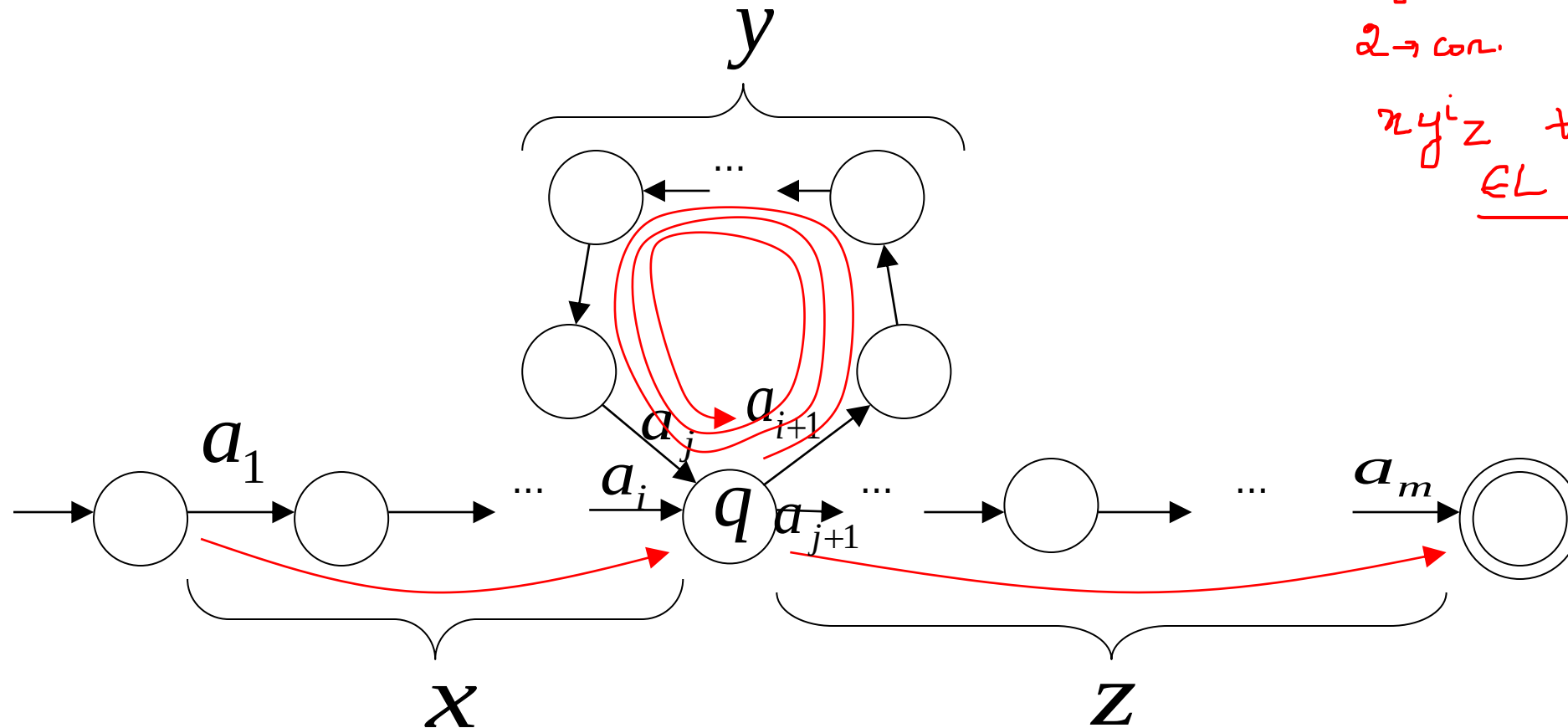
PUMPING LEMMA

- 3 occurrences of y (xy^3yz)



PUMPING LEMMA

- Many occurrences of y (xy^iz)
- xy^iz is also in L



$$w = xyz$$

$2 \rightarrow \text{con.}$

$$xy^iz \quad \forall i \geq 0$$

$\in L$

PUMPING LEMMA

- $\delta(q_0, a_1a_2\dots a_i) = q_i = q_j$
- $\delta(q_i, a_{i+1}\dots a_j) = q_i$, and
- $\delta(q_i, a_{j+1}\dots a_m) = q_n$

It is obvious that $\delta(q_i, y^i) = q_i$ for $i \geq 0$.

So, if the FA accepts $w = xyz$, it also accepts xy^iz .

APPLICATION

- Useful to prove a language L is not a regular set
- Method
 - Select an arbitrary 'n'
 - Choose a string w in L where $|w| \geq n$
 - For any partition of $w = xyz$ such that
 - $|xy| \leq n$ and $|y| \geq 1$, show a contradiction;
 - i.e. show that there is a string xy^kz not in L ;
 - k will depend on n , x , y , and z

EXAMPLE

- $L = \{a^m b^m / m \geq 1\}$

$$w = xyz \quad |w| \geq n$$

$$|xy| \leq n$$

$$\frac{|y| > 0}{y \neq \epsilon}$$

$$xyz \quad \forall i \geq 0 \quad \in L$$

$$a^m b^m / m \geq n$$

$$\sim n! \rightarrow$$

$$w = a^p b^p$$

$$|a^p b^p| \geq n$$

$$\frac{xyz}{xy = a^r}$$

$$y = a^q$$

$$z = a^{p-r} b^p$$

$$|a^r| \leq n$$

$$y \neq \epsilon$$

$$xy^i z = xy y^{i-1} z$$

$$xy^i z = a^r (a^q)^{i-1} a^{p-r} b^p$$

$$\underline{i=0} \quad a^r (a^q)^{-1} a^{p-r} b^p$$

$$= \cancel{a^r} a^{-q} a^{p-r} b^p$$

$$= a^{p-q} b^p \notin L$$

$$\underline{i=1} \quad a^r (a^q)^0 a^{p-r} b^p$$

$$= a^p b^p \in L$$

$$\underline{i=2} \quad a^r (a^q)^1 a^{p-r} b^p$$

$$= a^{p+q} b^p \notin L$$

$\therefore$ The language is not regular.

EXAMPLE

- $L = \{a^m / m \text{ is a prime}\}$

$w = a^p$ assume $|a^p| \geq n$
 $|w| \geq n$
 $xy = a^k$ where $|a^k| \leq n$
 assume $y = a^q$ $q \neq \varepsilon$
 $z = a^{p-k}$
 $\Rightarrow xy^iz$
 $xy^{i-1}z$
 $a^k (a^q)^{i-1} a^{p-k}$
 $i=0$
 $a^k a^{-q} a^{p-k}$ $a^{p-q} \notin L$ $p-q$ might not be prime
 $i=1$
 $a^k (a^q)^0 a^{p-k}$ $a^p \in L$
 $i=2$ p is a prime not accepted
 $a^k a^q a^{p-k}$ $a^{p+q} \notin L$ $p+q$ might not be prime

EXAMPLE

- $L = \{a^{i^2} \mid i \geq 1\}$
- a^i

SUMMARY

- Definition of Pumping lemma – Regular Language
- Application of pumping lemma

TEST YOUR KNOWLEDGE

- If we select a string w such that $w \in L$, and $w = xyz$. Which of the following portions cannot be an empty string?
 - a) x
 - b) y
 - c) z
 - d) all of the mentioned
- Which of the following one can relate to the given statement:
Statement: If n items are put into m containers, with $n > m$, then at least one container must contain more than one item.
 - a) Pumping lemma
 - b) Pigeon Hole principle
 - c) Count principle
 - d) None of the mentioned

REFERENCE

- Hopcroft J.E., Motwani R. and Ullman J.D, “Introduction to Automata Theory, Languages and Computations”, Second Edition, Pearson Education, 2008